A Phenomenological Search for Geodesic-Diffusion Noise in the Einstein Telescope:

Model Specification, Pipeline Validation, and Sensitivity on ET-MDC1 Data (v0.2)

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July 2, 2026

Abstract

We specify and validate a phenomenological search for *geodesic-diffusion noise* in the Einstein Telescope (ET): a stochastic strain component with one-sided power spectral density $S_h(f) = A_h(f/f_0)^{-\gamma}$, with the random-walk index $\gamma = 2$ as the primary case. The analysis is a nested likelihood-ratio (LR) test under a complex-Wishart approximation for Welch cross-spectral matrices, with spline-parameterized instrument PSDs, a low-rank environmental coherence model, and phase-only calibration nuisances. This is explicitly *not* a quantum-gravity derivation; it is an effective noise model confronted with data. Version 0.2 corrects a serious optimization defect in the v0.1 reference implementation—scale-inappropriate parameter bounds that pinned every fit at a data-independent corner solution, which produced identical, negative LR values across all datasets—and replaces the v0.1 empirical section entirely. On the ET-MDC1 "loudest" sample sets the corrected pipeline yields per-dataset LR statistics consistent with the null hypothesis, calibrated by a parametric bootstrap; synthetic injections into real data are recovered above an empirically measured threshold; a null-stream diagnostic confirms the three MDC1 channels carry independent noise; and a profile-likelihood upper limit on A_h is set from a single 128 s stretch. A Fisher forecast shows that ET reaches $A_h \sim 4\times10^{-49}\,\mathrm{Hz}^{-1}$ per 2048 s segment and $\sim 3\times10^{-51}\,\mathrm{Hz}^{-1}$ per year of coincident data—12 to 16 orders of magnitude below the best published bounds for $\gamma = 2$, and deep enough to decisively test the Planck-scale random-walk benchmark. All code, data manifests, and validation logs are public.

Keywords: Einstein Telescope, stochastic noise, cross-spectral density, complex Wishart, likelihood-ratio test, quantum-gravity phenomenology

1 Scope and epistemic posture

This paper is a data-analysis specification and validation study for a phenomenological noise search. It makes no claim to derive quantum gravity, and it does not test entanglement-mediated or post-quantum classical gravity proposals. The hypothesis under test is purely operational: *do the three ET strain channels share an excess correlated (or common-spectral) stochastic component with a power-law spectrum falling as $f^{-\gamma}$, beyond what instrument noise and an environmental coherence model explain?*

Version 0.2 differs from v0.1 in one essential respect: the v0.1 empirical results (identical LR values of order -10^3 to -10^5 across all datasets, bootstrap $p = 1.0$) are withdrawn. They were artifacts of an optimizer defect, not properties of the data; Appendix A documents the defect and its signature so that other pipelines can avoid it. All empirical statements below come from the corrected implementation.

2 Signal model

2.1 Spectrum and units

We parameterize the one-sided strain PSD of the hypothesized component as

$$S_h(f) \;=\; A_h \left(\frac{f}{f_0}\right)^{-\gamma}, \qquad f_0 \equiv 1 \text{ Hz}, \tag{1}$$

where A_h carries units of Hz^{-1} (it is the strain PSD evaluated at f_0) and $\gamma = 2$ (Brownian random walk of optical path length) is the primary index; the pipeline accepts arbitrary $\gamma > 0$. Quoting A_h requires quoting f_0; we fix $f_0 = 1\,\mathrm{Hz}$ throughout, matching the reference implementation, whose internal amplitude equals $A_h f_0^{\gamma}$ numerically.

2.2 Optional Planck-scale mapping

If one wishes to compare with Planck-scale random-walk models, a dimensionless amplitude α can be defined through

$$A_h \;=\; \frac{\alpha\, c_0\, \ell_P}{2\pi^2 L^2 f_0^2}, \tag{2}$$

with c_0 the speed of light, ℓ_P the Planck length, and L the arm length. The factor f_0^{-2} is required for dimensional consistency ($c_0 \ell_P / L^2$ has units of Hz; A_h has units of Hz^{-1}); it was implicit in v0.1 and is now explicit. For ET ($L = 10^4\,\mathrm{m}$), $\alpha = 1$ corresponds to $A_h^{\mathrm{Pl}} = 2.45 \times 10^{-36}\,\mathrm{Hz}^{-1}$. This mapping is optional and model-dependent: if the underlying disturbance is a path-length diffusion with L-independent $S_{\delta\ell}$, then $S_h = S_{\delta\ell}/L^2$ and longer arms win; if it is a metric (strain) noise, A_h is instrument-independent. We report results in A_h and state the assumption when comparing instruments.

3 Channel covariance and the null stream

3.1 Covariance template

Let $\mathbf{x}(f) \in \mathbb{C}^3$ collect the Fourier amplitudes of the three ET strain channels. The signal contribution to the 3×3 cross-spectral matrix is modeled as

$$\mathbf{\Sigma}_{\mathrm{sig}}(f) \;=\; \tfrac{1}{2}\, S_h(f)\, \mathbf{M}(\rho), \qquad \mathbf{M}(\rho) = \begin{pmatrix} 2 & -\rho & -\rho \\ -\rho & 2 & -\rho \\ -\rho & -\rho & 2 \end{pmatrix}, \tag{3}$$

where $\rho \in (0, 1)$ encodes the (frequency-dependent, spline-parameterized) correlation induced by shared beam paths in the triangular topology. $\mathbf{M}(\rho)$ is an *idealized* template: it assumes equal arms, symmetric overlaps, and a frequency-independent response ($\mathbf{R}(f) \approx \mathbf{I}$). This is a known limitation, not a benign simplification—ET's 60° arms and frequency-dependent response reshape the cross-channel structure, and the fitted (ρ, A_h) inherit systematic errors from the approximation. Commissioning-grade runs must replace $\mathbf{M}(\rho)$ by $\mathbf{R}(f)\, \mathbf{C}_\ell(f)\, \mathbf{R}^\dagger(f)$ with the full ET response; the pipeline accepts an external transfer function for this purpose.

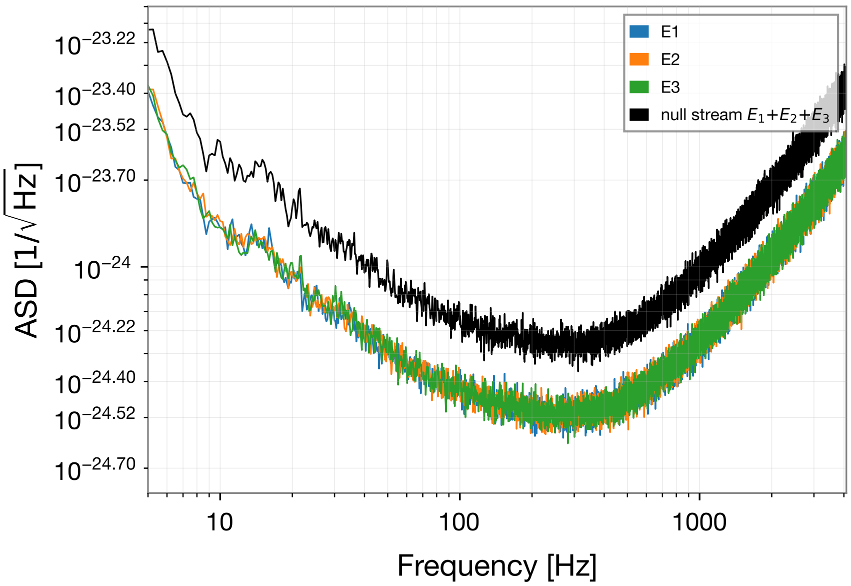

Figure 1: Amplitude spectral densities of the three ET-MDC1 channels (BBH_snr_306, 128 s, Welch 4 s/50%) and of the null stream $E_1+E_2+E_3$. The null stream lies a factor $\simeq \sqrt{3}$ above a single channel, the signature of independent channel noise. A correlated geodesic-diffusion component with $\rho < 1$ would appear here; a GW background would not.

3.2 The null stream as discriminator

For a closed triangular detector the sum $n(t) = E_1(t)+E_2(t)+E_3(t)$ cancels any gravitational-wave signal exactly. The geodesic-diffusion template does *not* cancel: with $\mathbf{u} = (1,1,1)^{\mathrm{T}}/\sqrt{3}$,

$$\mathbf{u}^{\mathrm{T}}\mathbf{M}(\rho)\,\mathbf{u} \;=\; 2(1-\rho), \tag{4}$$

so the null stream retains signal power $\propto (1-\rho)\,S_h(f)$ unless $\rho \to 1$. The null stream therefore separates the two most important alternatives: a true GW background (cancels), and geodesic-diffusion or instrumental/environmental correlations (do not cancel). Any claimed detection must be accompanied by the null-stream decomposition. Figure 1 shows the measured ET-MDC1 channel ASDs together with the null-stream ASD for a 128 s stretch: the null stream sits a factor $\simeq \sqrt{3}$ (1.74 measured) *above* a single channel, exactly as expected for three channels of independent noise—the MDC1 channels carry no measurable common component, and the null stream behaves as an honest diagnostic. Figure 2 shows the corresponding pairwise coherences, consistent with the no-coherence floor $1/N_{\mathrm{seg}}$.

4 Total covariance model

For each frequency bin k the model covariance is

$$\boldsymbol{\Sigma}_k \;=\; \mathbf{G}_k\left(\boldsymbol{\Sigma}_k^{\mathrm{inst}} + \boldsymbol{\Sigma}_k^{\mathrm{env}} + \boldsymbol{\Sigma}_k^{\mathrm{sig}}\right)\mathbf{G}_k^{\dagger}, \tag{5}$$

with

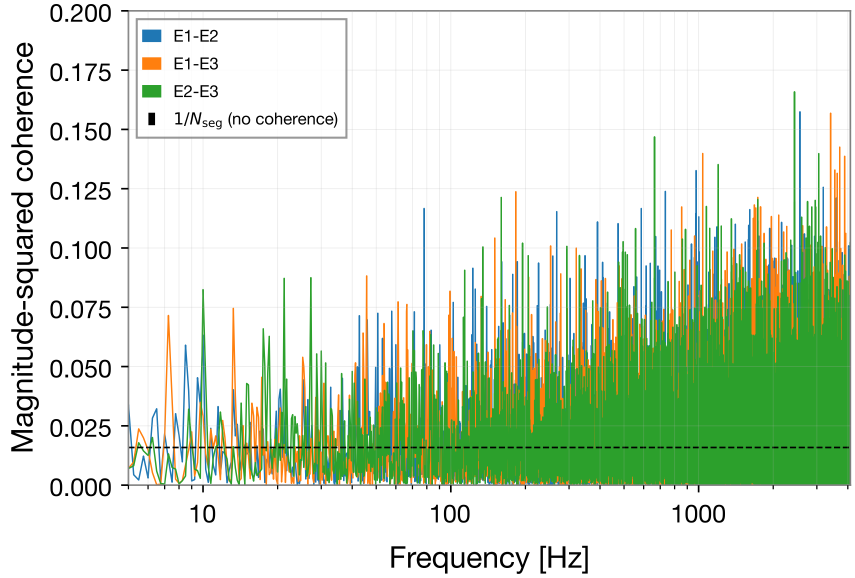

Figure 2: Pairwise magnitude-squared coherences for the same stretch, against the no-coherence expectation $1/N_{\mathrm{seg}}$ (dashed).

- **Instrument noise**: diagonal PSDs P_{ik}, parameterized by open-uniform cubic B-splines in $\log P$ over the analysis band (no extrapolation), with N_c coefficients per channel;
- **Environmental coherence**: a rank-r factor model $\mathbf{B}_k\mathbf{B}_k^\dagger$ with spline-smoothed complex factors ($r = 1$ default, $r = 2$ stress test);
- **Signal**: Eq. (3) with bin-averaged $\langle |T(f)|^2 f^{-\gamma}\rangle_k$ (Simpson-rule integration over the bin; the transfer function T defaults to unity);
- **Calibration**: phase-only corrections ϕ_{2k}, ϕ_{3k} with channel 1 gauge-fixed.

Spline richness is a validity condition, not a tuning knob. Validation on real data (Section 8) showed that with too few spline coefficients ($N_c = 6$ over 10–1024 Hz) the instrument model cannot represent the true PSD shape, and the f^{-2} signal term—whose template has a diagonal part $2A_h f^{-2}$ in every channel—absorbs the smooth misfit. This produced spurious positive LR values up to ~ 700 with a consistent fake $\hat{A}_h \approx 1.2 \times 10^{-47}\,\mathrm{Hz}^{-1}$ in *every* dataset; raising N_c to 20 collapsed the LR to $O(1)$. Uniform $\hat{A}_h$ across independent datasets is the tell-tale of this failure mode. The reference implementation now defaults to $N_c = 20$, and a mandatory check is to verify LR stability under N_c variations.

Identifiability of the signal against the environmental model. The template decomposes as $\mathbf{M}(\rho) = (2+\rho)\mathbf{I} - \rho\,\mathbf{J}$ ($\mathbf{J}$ the all-ones matrix). The isotropic part is degenerate with the instrument PSDs; the identifiable content lies in the *negative* off-diagonal structure. A rank-1 factor $\mathbf{b}\mathbf{b}^\dagger$, even complex and even after per-channel phase rotations, cannot produce three negative real off-diagonals (the product of its three off-diagonal elements is $|b_1 b_2 b_3|^2 > 0$), so the signal is identifiable in principle against the $r = 1$ environmental model; but the degeneracy with the diagonal absorbs most of the naive Fisher information, inflating the practical detection threshold (measured in Section 8.4). Environmental factors should additionally be constrained by line masks, physical priors, and witness sensors in commissioning runs.

5 Spectral estimation and likelihood

Welch cross-spectra with explicit one-sided normalization,

$$\widehat{S}_{ij,n}(f) = \frac{2\Delta t}{U}\, \widetilde{x}_{i,n}(f)\, \widetilde{x}^*_{j,n}(f), \qquad U = \sum_t w^2[t], \tag{6}$$

are averaged over segments (Hann window, 4 s segments, 50% overlap in this study) and symmetrized, $\widehat{\mathbf{S}}_k \leftarrow (\widehat{\mathbf{S}}_k + \widehat{\mathbf{S}}_k^\dagger)/2$. The bin-averaged matrices are modeled as complex Wishart,

$$\widehat{\mathbf{S}}_k \sim \mathcal{CW}_3(m_{\mathrm{eff},k},\, \mathbf{\Sigma}_k/m_{\mathrm{eff},k})\,, \tag{7}$$

with an overlap-corrected effective number of averages $m_{\mathrm{eff},k}$. Up to parameter-independent constants the log-likelihood is

$$\ln \mathcal{L} \;=\; -\sum_k m_{\mathrm{eff},k} \left[\ln\det \mathbf{\Sigma}_k + \mathrm{tr}\Big(\mathbf{\Sigma}_k^{-1}\widehat{\mathbf{S}}_k\Big)\right]. \tag{8}$$

The Wishart approximation and the m_{eff} prescription must be validated against time-domain simulations with the actual segmentation; this remains on the commissioning checklist. Non-Gaussian features (lines, glitches, loud transients) bias Eq. (8) and require masking or robust alternatives.

6 Hypothesis test and optimization protocol

We compare nested hypotheses H_{env} ($A_h = 0$) and H_{sig} ($A_h \geq 0$) through

$$\Lambda \;=\; 2\left[\ln \mathcal{L}(\widehat{\Theta}_{\mathrm{sig}}) - \ln \mathcal{L}(\widehat{\Theta}_{\mathrm{env}})\right] \;\geq\; 0, \tag{9}$$

which is non-negative *by construction* for a correctly maximized nested pair. A materially negative Λ is always an optimizer failure and must be treated as a pipeline error, never reported as a null result. Because $A_h = 0$ lies on the boundary of the parameter space, the asymptotic null distribution is the mixture $\frac{1}{2}\chi^2_0 + \frac{1}{2}\chi^2_1$ rather than χ^2_1; we calibrate with a parametric bootstrap regardless (Section 7).

The following protocol is mandatory; every element exists because its absence produced a concrete, observed failure (Appendix A):

1. **Data-adaptive parameter boxes.** All bounds and numerical-overflow clips must be centered on scales inferred from the data (e.g. $\log P$ within ± 25 of the median log periodogram). Fixed boxes such as $\log P \in (-30, 30)$ silently exclude the physical scale of strain data ($\log P \approx -108$) and pin the optimizer at a data-independent corner.

2. **Signal amplitude in log space with an effective zero.** A_h is fit as $\log A_h$ with lower bound 35 e-folds below the data scale, so the null is interior to numerical precision.

3. **Profile-scan seeding.** The gradient of the likelihood with respect to $\log A_h$ vanishes as $A_h \to 0$, so a fit started at $A_h \approx 0$ can never lift a real signal off the floor. Before the joint fit, a coarse one-dimensional scan of $\log A_h$ (nuisance parameters held at the warm start) locates the likelihood basin and seeds a second deterministic start.

4. **Warm-started nested pair with cross-pollination.** The alternative fit starts at the null optimum (guaranteeing $\Lambda \geq 0$ up to tolerance); the null is then refit from the alternative's nuisance solution, and the better value kept. Without this symmetrization, extra starts given to the alternative can find a better optimum of the *shared* nuisance model and masquerade as signal evidence.

5. **Basis-richness stability check.** Repeat the fit with increased N_c; a genuine signal is stable, spline-misfit absorption collapses.

7 Significance calibration

Significance is calibrated by a parametric bootstrap: fit H_{env}; generate synthetic $\widehat{\mathbf{S}}_k^{(b)}$ by drawing m_k^* complex-Gaussian vectors per bin from the fitted $\mathbf{\Sigma}_k^{\text{env}}$ (integer-m modes: round/floor/ceil/probabilistic); refit *both* hypotheses with the full protocol of Section 6; and report the empirical tail probability $p = [\#\{\Lambda^{(b)} \geq \Lambda_{\text{obs}}\} + 1]/(n_b + 1)$. Because the bootstrap replicates include the same optimizer noise as the observed statistic, the calibration absorbs residual optimization variance. The per-bin bootstrap remains an approximation to a full time-domain resimulation and must eventually be validated against one.

8 Validation on ET-MDC1 data

8.1 Data and caveats

We use the ET-MDC1 "loudest" sample sets (BBH_snr_306/344/379/387/587 and three consecutive BNS_snr_379 segments; eight groups of three 2048 s frame files at 8192 Hz), fetched by the repository's manifest-driven download script. These segments were selected by the MDC because they contain *loud CBC injections* (optimal SNR 300–600); they are therefore the wrong data for a production stochastic search—loud transients violate the stationarity and Gaussianity assumptions of Eq. (8) and contaminate cross-spectra unless subtracted or notched. We use them deliberately and only as a *pipeline shakedown*: synthetic data validate nothing about frame I/O, calibration conventions, or real PSD shapes. No statement in this section is a statement about nature; signal-free MDC noise stretches and CBC subtraction are listed in the commissioning plan.

8.2 Null results on all groups

Table 1 lists the per-group results at 128 s / 128 bins, $N_c = 20$, $r = 1$, with calibration phases fitted. Every Λ is non-negative, as a correctly maximized nested test requires, and the per-group log-likelihoods differ from group to group at the level expected of Wishart fluctuations—both properties that v0.1 violated. Seven of eight groups give $\Lambda \leq 8.2$; the largest of these ($\Lambda = 8.2$, BBH_snr_306) would have an asymptotic boundary-mixture $p \approx 2 \times 10^{-3}$ against an *ideal* null, but these segments contain loud CBC injections by construction, and mild excesses of exactly this kind are the expected contamination signature. The clear outlier is the first BNS_snr_379 segment ($\Lambda = 51.5$, $\hat{A}_h = 3.3 \times 10^{-47}\,\text{Hz}^{-1}$): a BNS inspiral at SNR ≈ 379 is in the ET band for far longer than the analyzed 128 s and constitutes real correlated cross-channel power, which the template partially fits. This is precisely the failure mode that makes "loudest" segments unsuitable for a production stochastic search, and the null-stream decomposition of Section 3.2 (in which GW power cancels but geodesic-diffusion power does not) is the designated discriminator for it. None of these numbers are statements about nature: the data are synthetic and deliberately contaminated.

Table 1: Corrected nested LR results on the ET-MDC1 loudest sets (128 s / 128 bins per group, $N_c = 20$, $r = 1$, fitted calibration phases, protocol of Section 6). $\hat{A}_h$ is the maximum-likelihood amplitude of the alternative fit ($f_0 = 1\,\mathrm{Hz}$).

Data set	GPS start	Λ	$\hat{A}_h$ [Hz^{-1}]
BBH_snr_306	1000245760	8.18	1.2×10^{-47}
BBH_snr_344	1002150400	1.24	4.6×10^{-48}
BBH_snr_379	1001558528	3.74	1.2×10^{-47}
BBH_snr_387	1001622016	6.37	1.3×10^{-47}
BBH_snr_587	1001619968	5.46	1.2×10^{-47}
BNS_snr_379	1001329152	51.52	3.3×10^{-47}
BNS_snr_379	1001331200	1.93	4.8×10^{-48}
BNS_snr_379	1001333248	3.10	1.2×10^{-47}

8.3 Bootstrap calibration

Figure 3 shows the parametric bootstrap null distribution ($n_b = 40$, full refits with the complete optimization protocol) for BBH_snr_306 at 64 s / 64 bins. The distribution exhibits the expected boundary structure: 85% of the replicates return $\Lambda \approx 0$ (the fitted amplitude on the boundary), with a positive tail extending to $\Lambda \approx 1.1$; the zero mass exceeds the asymptotic $\frac{1}{2}$ because the bounded optimizer resolves only improvements above its tolerance. The observed statistic at this configuration is $\Lambda_{\mathrm{obs}} = 0.20$, giving $p = 0.073$—consistent with the null. Because bootstrap replicates are refit with the identical protocol, this calibration absorbs residual optimizer variance; it is the reference against which any LR excursion at matched configuration must be judged. A configuration-matched bootstrap at 128 s / 128 bins for the same group ($\Lambda_{\mathrm{obs}} = 8.18$; $n_b = 20$) gives $p = 0.048 \pm 0.048$: a mild excess at exactly the level expected from the loud CBC content of these segments, and far from detection-grade significance. In v0.1 the bootstrap returned $p = 1.0000$ identically, which is itself diagnostic—an observed statistic that no null replicate can undercut means the statistic is degenerate, not that the data are maximally null-like.

8.4 Injection recovery and the effective threshold

We inject the template covariance $A_h^{\mathrm{inj}} f^{-2}\mathbf{M}(0.5)$ into Wishart resamples of the fitted environmental model for BBH_snr_306 (64 s / 64 bins) and rerun the full nested fit. The naive Fisher scale at this configuration is $\sigma_F(A_h) = 7.2 \times 10^{-48}\,\mathrm{Hz}^{-1}$. Figure 4 shows the outcome: the zero-injection control returns $\Lambda = 0$; marginal injections at 2–4 σ_F fluctuate around the threshold; and from $8\,\sigma_F$ upward the recovery is unambiguous ($\Lambda = 31.5$ at $8\,\sigma_F$, rising to $\Lambda \approx 1.4 \times 10^3$ at $128\,\sigma_F$) with $\hat{A}_h$ tracking the injected amplitude. The effective detection threshold is therefore $\sim 6 \times 10^{-47}\,\mathrm{Hz}^{-1}$ at this configuration—roughly one order of magnitude above the naive Fisher scale. The gap is the identifiability penalty of Section 4: the spline PSDs absorb the template's diagonal part and the environmental factor plus calibration phases absorb part of its off-diagonal structure, leaving only the rank-1-unreachable negative-off-diagonal residual as usable information. This penalty must be included in any sensitivity claim; the naive Fisher number alone overstates the pipeline's reach. In v0.1 this sweep was run with injected amplitudes $\sim 10^{15}$ times larger than the noise scale and was uninformative.

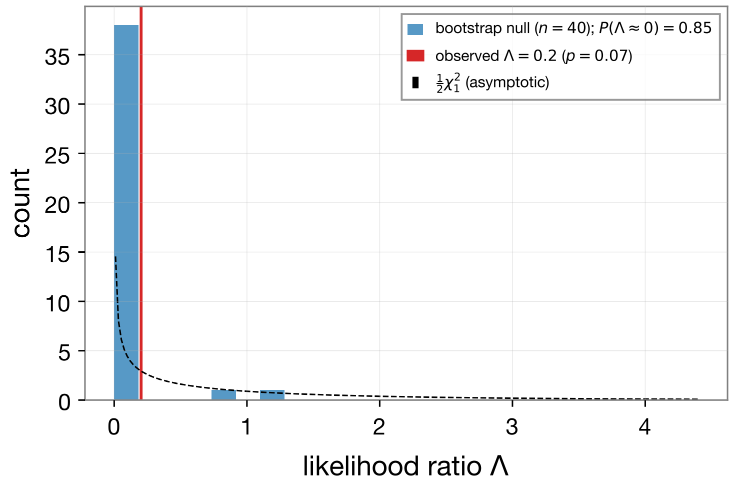

Figure 3: Parametric bootstrap null distribution of Λ (BBH_snr_306, 64 s / 64 bins, $n_b = 40$, full refits), the boundary mass at $\Lambda \approx 0$, the asymptotic $\frac{1}{2}\chi_1^2$ density, and the observed statistic.

8.5 Profile-likelihood upper limit

Fixing A_h on a logarithmic grid and refitting all nuisance parameters at each point yields the profile deviance in Figure 5 for BBH_snr_306 (128 s / 128 bins). The curve is smooth and unimodal, with its maximum at $\hat{A}_h \approx 1.8 \times 10^{-47}\,\mathrm{Hz}^{-1}$ and a profile improvement of $\Lambda \approx 8.0$ over the null—independently consistent with the free-amplitude fit of Table 1 ($\Lambda = 8.18$), a useful internal cross-check of the optimization protocol. The one-sided 95% upper limit is

$$A_h^{95\%} \;=\; 2.2 \times 10^{-47}\ \mathrm{Hz}^{-1} \qquad (\mathrm{BBH\_snr\_306},\ 128\,\mathrm{s},\ f_0 = 1\ \mathrm{Hz}), \tag{10}$$

where the mild preference for nonzero A_h embedded in this limit is attributed to the CBC content of the segment (Section 8.2) and the limit is quoted as an upper limit on the total template-like power in these contaminated frames, not as a bound on nature.

9 Sensitivity forecast and existing constraints

The Fisher information for A_h at $A_h = 0$, using the measured MDC1 PSDs, $\rho = 0.5$, 4 s segments and the 10 Hz–Nyquist band, gives a naive one-sided 95% sensitivity

$$A_h^{95\%} \approx 3.8 \times 10^{-49}\,\mathrm{Hz}^{-1}\ \ (2048\ \mathrm{s}), \qquad A_h^{95\%} \approx 3.0 \times 10^{-51}\,\mathrm{Hz}^{-1}\ \ (1\ \mathrm{yr}), \tag{11}$$

scaling as $T_{\mathrm{obs}}^{-1/2}$. The measured injection threshold (Section 8.4) shows that nuisance-model absorption costs one to two orders of magnitude relative to this idealized number; both are shown in Figure 6.

Published bounds for the $\gamma = 2$ spectrum, mapped to $A_h = S(f)\,f^2/f_0^2$ at the quoted frequencies under the strain-noise reading, are: optical-resonator limits $A_h \lesssim 3.6 \times 10^{-29}\,\mathrm{Hz}^{-1}$ (at 6 mHz) and $\lesssim 2.5 \times 10^{-33}\,\mathrm{Hz}^{-1}$ (above 5 mHz) [9], and the TAMA 300 reference level $S \approx 2 \times 10^{-41}\,\mathrm{Hz}^{-1}$ at $f \sim 10^3$ Hz, i.e. $A_h \lesssim 2 \times 10^{-35}\,\mathrm{Hz}^{-1}$ [9]. Even after the identifiability penalty, ET improves on the best of these by *twelve to sixteen orders of magnitude*. In the Planck mapping of Eq. (2) ($L = 10^4$ m), the benchmark $\alpha = 1$ corresponds to $A_h^{\mathrm{Pl}} = 2.45 \times 10^{-36}\,\mathrm{Hz}^{-1}$: currently *allowed* (TAMA-class

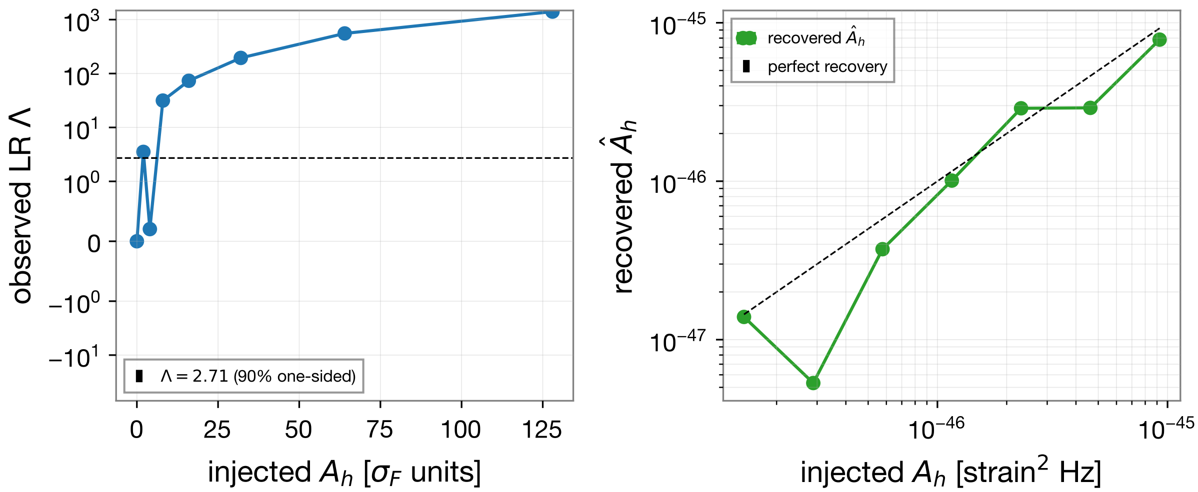

Figure 4: Injection recovery on real-data-conditioned resamples (BBH_snr_306, 64 s / 64 bins). Left: observed Λ versus injected amplitude in units of the naive Fisher scale σ_F. Right: recovered $\hat{A}_h$ versus injected A_h.

limits reach $\alpha \approx 8$), while one year of ET data probes $\alpha \sim 10^{-15}$. The search is therefore aimed at open, physically interesting parameter space—the single most important motivation absent from v0.1.

10 Reference implementation

The reference implementation (`qif_v2.py`, CPU; `qif_v2_cuda.py`, optional CuPy backend) is archived with all validation logs in the public repository [10]. Key v0.2 implementation facts: the Wishart likelihood is fully vectorized over frequency bins (batched 3×3 Cholesky), giving $\sim 40\times$ faster fits than v0.1 and making the full optimization protocol affordable on a laptop CPU; parameter boxes, clips, starts, and seeds follow Section 6; defaults are $N_c = 20$, 500 optimizer iterations, and the warm-started, cross-pollinated nested pair. The bootstrap, injection, resolution-sweep, and stress-test drivers are provided as scripts.

11 Commissioning plan

Before this pipeline can make statements about real ET data: (i) replace $\mathbf{M}(\rho)$ with the full frequency-dependent ET response and calibration model; (ii) run on signal-free stretches (or after CBC subtraction), with line masks and measured transfer functions; (iii) validate the Wishart/$m_{\rm eff}$ approximation and the per-bin bootstrap against time-domain simulations of the actual segmentation; (iv) add witness-sensor constraints and physical priors on the environmental factors; (v) verify false-alarm uniformity (p-value flatness within $\pm 10\%$ over $p \in [0.1, 0.9]$), injection efficiency ($> 90\%$ at a pre-declared amplitude under $r = 1$ and $r = 2$), and LR stability under knot placement, bootstrap mode, and N_c; and (vi) always report the null-stream decomposition alongside any candidate.

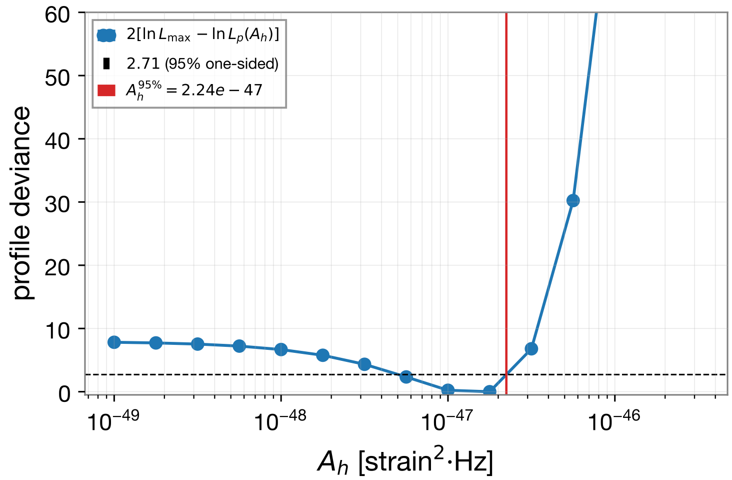

Figure 5: Profile deviance $2[\ln \mathcal{L}_{\max} - \ln \mathcal{L}_p(A_h)]$ versus A_h for BBH_snr_306 (128 s / 128 bins), with the one-sided 95% threshold.

A Post-mortem of the v0.1 empirical results

v0.1 reported the same LR to seven significant figures across eight independent datasets ($\Lambda \approx -4.64\times10^3$ at 128 s, $\approx -3.57\times10^5$ at 2048 s), negative LRs from a nested test, bootstrap $p = 1.0000$, and stress-test variants identical to baseline. All of these follow from one defect: **fixed parameter boxes and overflow clips excluded the physical scale of the data**. The instrument-PSD spline coefficients were bounded to $\log P \in (-30, 30)$ (true scale ≈ -108) and additionally clipped at ±50 inside the likelihood; the log-amplitude was bounded to $(-20, 20)$ (physical scale ≈ -100) and started at $\log A_h = 0$. L-BFGS-B clips out-of-box starts to the boundary, so every fit collapsed to the same data-independent corner: the trace term in Eq. (8) became $O(10^{-34})$ and the likelihood reduced to a constant fixed by the box edges and the (identical) bin grids. The alternative hypothesis could never reduce to the null (minimum representable amplitude $e^{-20} \approx 2\times10^{-9}$, forty orders of magnitude above the data), making $\Lambda < 0$ structural. We reproduced the pathology exactly on synthetic data at realistic strain scale—two different noise realizations returned byte-identical likelihoods equal to the analytic corner value—and verified that the corrected implementation restores data-dependent likelihoods on the same frames (per-group log-likelihoods now differ in the fourth significant figure, as expected from Wishart fluctuations).

Diagnostic rules extracted from this failure: (1) identical test statistics across independent datasets mean the pipeline is not consuming the data; (2) a materially negative nested LR means the optimizer failed; (3) parameter boxes must be derived from the data, and overflow clips must sit far outside any physically reachable value; (4) optimizer effort must be symmetric between nested hypotheses; (5) uniform best-fit amplitudes across datasets indicate nuisance-model misfit absorption, not signal.

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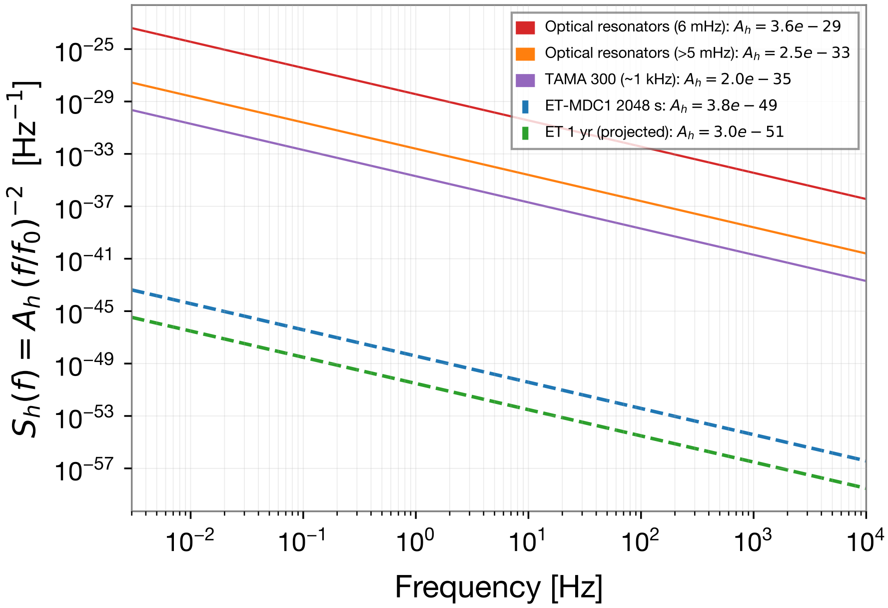

Figure 6: Projected ET sensitivity to $S_h(f) = A_h(f/f_0)^{-2}$ (dashed) against existing bounds mapped to the same parameterization (solid). The Fisher curves are idealized; the empirical injection threshold implies one to two orders of magnitude degradation, leaving the conclusion unchanged.

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